# PROJECT REPORT


## Abstract

This study compares two traditional machine-learning classifiers with a transformer-based classifier for email Spam/Ham classification. I have seen that the traditional pipeline uses Term Frequency–Inverse Document Frequency (TF-IDF) features with Logistic Regression and Multinomial Naive Bayes. The transformer approaches fine-tune DistilBERT using PyTorch and a custom training loop, and the Hugging Face Trainer API is not used. The cleaned dataset contained 5,157 messages. On the held-out test set, DistilBERT achieved 98.74% accuracy and a Spam F1-score of 94.74%. The traditional classifiers trained substantially faster but detected fewer Spam messages.


## 1. Introduction

Email spam classification is the job of separating potentially harmful messages from legitimate email, which we call Ham. In my view Email spam classification traditionally uses word-based features and trains classifiers. Transformer models learn contextual text representations during pretraining and can be fine-tuned for classification. This project compares TF-IDF with Logistic Regression, TF-IDF with Multinomial Naive Bayes, and a fine-tuned DistilBERT model using the same held-out test set.

## 2. Dataset and Preprocessing

The dataset was obtained from Kaggle: Spam Email Classification (https://www.kaggle.com/datasets/ashfakyeafi/spam-email-classification). The columns were Message (email text) and Category (label). The downloaded dataset contained 5,573 rows and two columns. The inspected columns had no missing values, and one unusual category value, {"mode": full"}, was observed. The cleaning code retained non-empty messages with labels ham or spam, removed duplicate message text, and mapped Ham to 0 and Spam to 1. The final cleaned dataset contained 5,157 messages: 4,516 Ham (87.57%) and 641 Spam (12.43%). This imbalance matters because a model can achieve high accuracy while still missing Spam.

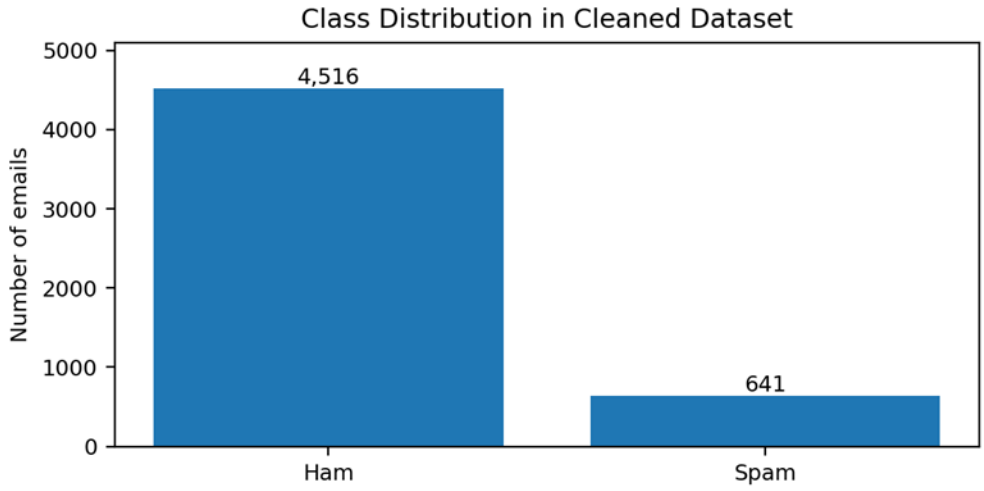



Figure 1. Class distribution in the cleaned dataset.

## 3. Methodology

A stratified split produced 3,712 training messages (72%), 413 validation messages (8%), and 1,032 test messages (20%). The test set contained 904 Ham and 128 Spam messages. Validation data was used to monitor DistilBERT; the held-out test set was used for final comparison.

### 3.1 Traditional machine learning

Email text was lowercased and whitespace normalized. TF-IDF uses unigrams and bigrams, a maximum vocabulary of 50,000 features, minimum document frequency 2, sublinear term frequency, and Unicode accent stripping. The vectorizer was fitted only on training data to avoid test data leakage. Logistic Regression used max_iter=1000; Multinomial Naive Bayes used alpha=1.0. Both models used the same TF-IDF features and test examples.

### 3.2 DistilBERT

The pretrained distilbert/distilbert-based-uncased model was fine-tuned for two classes using PyTorch. Settings were maximum sequence length 128, batch size 16, three epochs, learning rate 2 × 10■■, AdamW optimizer, weight decay 0.01, a linear learning-rate schedule with warm-up, and gradient clipping at 1.0. All model layers were trainable. A custom training loop performed forward passes, loss calculation, backpropagation, optimizer updates, and validation; the Hugging Face Trainer API was not used.

### 3.3 Evaluation metrics

Accuracy is the fraction of all predictions that are correct. Spam precision measures how many messages predicted as Spam are truly Spam. Spam recall measures how many real Spam messages are detected. Spam F1-score balances precision and recall. Confusion matrices show actual versus predicted classes. Training time was measured during model fitting in the Collab runtime.

## 4. Experimental Setup

The experiment was conducted using Google Colab with a CUDA-enabled GPU. The dataset was split into 72% training, 8% validation, and 20% testing using stratified sampling. Logistic Regression and Multinomial Naive Bayes were trained using TF-IDF features, while DistilBERT was fine-tuned using PyTorch for 3 epochs with a batch size of 16 and a learning rate of $2 \times 10^{-5}$. All models were evaluated using accuracy, precision, recall, F1-score, confusion matrices, and training time.

## 5. Results

Table 1 reports the metrics shown in the completed Colab notebook. Spam is the positive class for precision, recall, and F1-score. DistilBERT achieved the highest accuracy (98.74%), recall (91.41%), and F1-score (94.74%). The traditional models both had 100% Spam precision in this run, but lower recall.

| Model | Accuracy | Precision | Recall | F1-score | Training time |
|---|---|---|---|---|---|
| Logistic Regression | 0.9622 | 1.0000 | 0.6953 | 0.8203 | 0.0416 s |
| Multinomial Naive Bayes | 0.9612 | 1.0000 | 0.6875 | 0.8148 | 0.0031 s |
| DistilBERT | 0.9874 | 0.9832 | 0.9141 | 0.9474 | 131.7066 s |

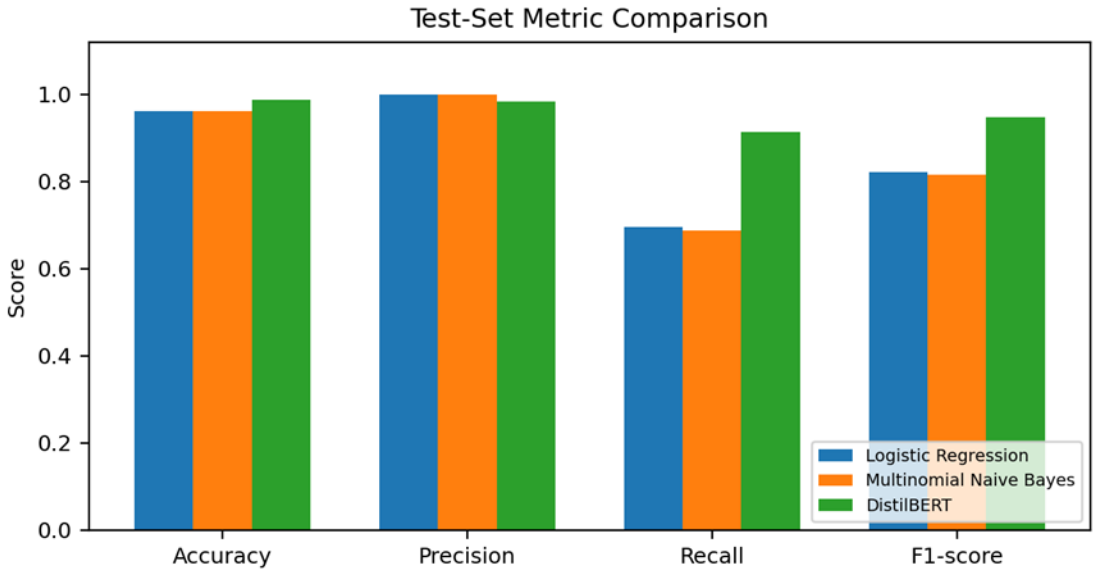



Figure 2. Test-set metric comparison.

## 6. Discussion

Logistic Regression correctly classified all 904 Ham messages and detected 89 of 128 Spam messages, missing 39. Naive Bayes correctly classified all 904 Ham messages and detected 88 Spam messages, missing 40. DistilBERT correctly classified 902 Ham and 117 Spam messages, with 2 Ham messages flagged as Spam and 11 missed Spam messages. Thus, DistilBERT detected substantially more Spam while introducing a small number of false positives.

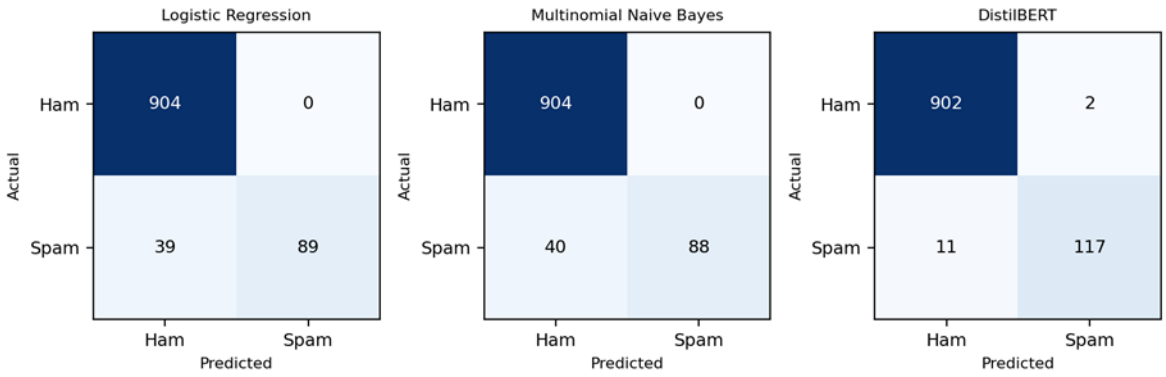



Figure 3. Confusion matrices (rows: actual; columns: predicted).

## 6.1 Computational cost and training behavior

Measured training times were 0.0416 seconds for Logistic Regression, 0.0031 seconds for Naive Bayes, and 131.7066 seconds for DistilBERT. Shared TF-IDF fitting took about 0.19 seconds. These times depend on the runtime and measurement method and are not universal benchmarks. The submitted Colab plot shows training and validation losses decreasing across three epochs, while validation Spam F1 remained nearly constant and close to 1.0. The held-out test Spam F1 was 0.9474, so conclusions are based on the independent test metrics rather than validation alone.

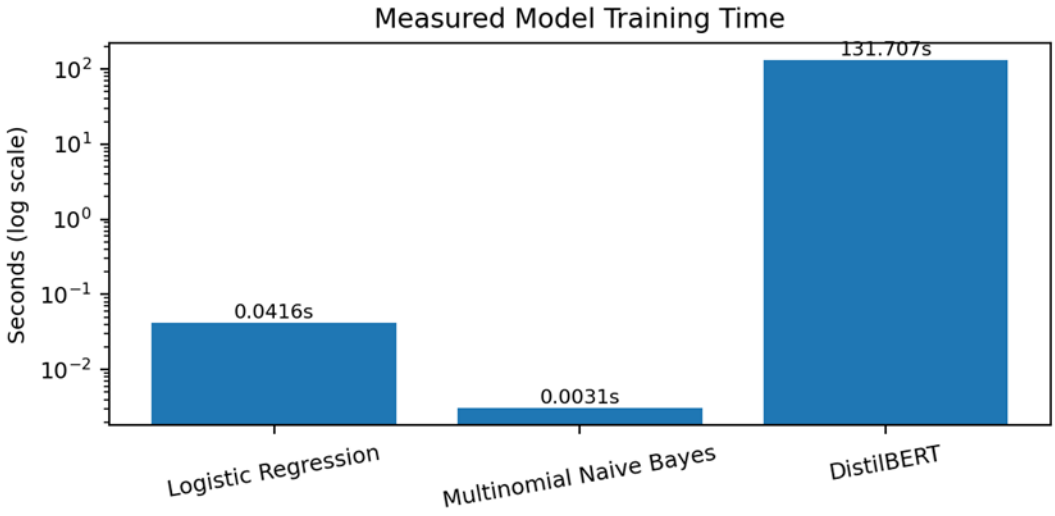



Figure 4. Measured model training time; logarithmic axis

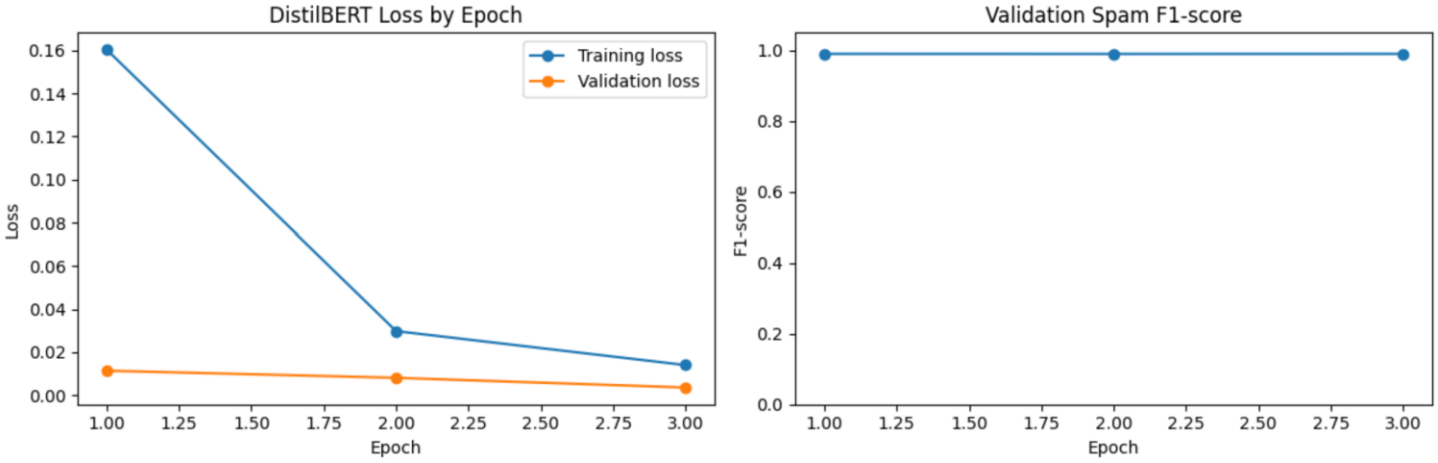



Figure 5. DistilBERT training curves as displayed in the submitted Collab output.

## 7. Limitations

1. Limited Dataset: In this experiment we used one dataset and a single train-test split. Because of that, the results might not work well with email sources or fresh spam messages.Class Imbalance: Spam accounts for only 12.43% of the dataset, which may affect model performance.
2. High Computational Cost: DistilBERT requires more processing power and training time than traditional machine learning models.
3. Text Truncation: DistilBERT processes a maximum of 128 tokens, so some information in longer emails may be lost.
4. Limited Hyperparameter Tuning: Only one main configuration and three training epochs were tested.
5. Generalization: We need to test the models on more datasets and run repeated train-test splits to confirm that the models are reliable.
6. Validation–Test Difference: DistilBERT's validation F1-score is different, from its test F1-score, which shows how important it is to evaluate the model on unseen test data.

## 8. Conclusion

On this test set, DistilBERT performed best overall, achieving 98.74% accuracy, 91.41% Spam recall, and a 94.74% Spam F1 score. Traditional models trained much faster and had perfect Spam precision in this run but missed more Spam messages. DistilBERT is preferable when detecting more Spam is more important than computational cost; traditional models remain useful for lightweight baselines.

## References

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https://www.kaggle.com/datasets/ashfakyeafi/spam-email-classification

2. Hugging Face. DistilBERT documentation.

https://huggingface.co/docs/transformers/en/model_doc/distilbert

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https://scikit-learn.org/stable/modules/generated/sklearn.feature_extraction.text.TfidfVectorizer.html

4. Scikit-learn. Classification metrics documentation.
https://scikitlearn.org/stable/modules/model_evaluation.html#classification-metrics

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